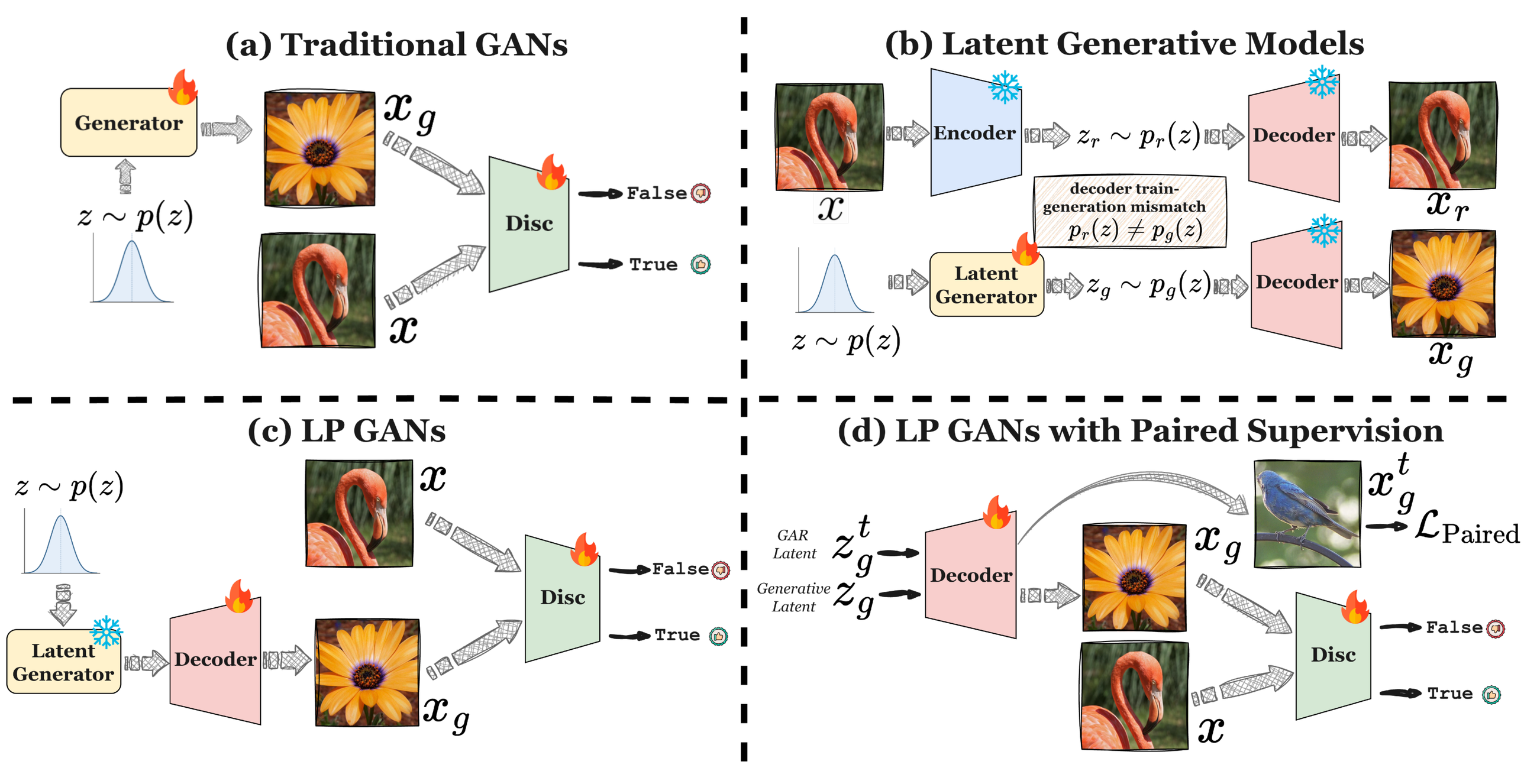

(a) Traditional GANs
Generator
$z \sim p(z)$
x_g
x
Disc
False
True
(b) Latent Generative Models
Encoder
$z_r \sim p_r(z)$
Decoder
x_r
x
decoder train-generation mismatch
$p_r(z) \neq p_g(z)$
Latent Generator
$z_g \sim p_g(z)$
Decoder
x_g
$z \sim p(z)$
(c) LP GANs
$z \sim p(z)$
x
Latent Generator
Decoder
x_g
Disc
False
True
(d) LP GANs with Paired Supervision
GAR Latent
Generative Latent
z_g^t
z_g
Decoder
x_g
x_g^t
$\mathcal{L}_{\text{Paired}}$
x
Disc
False
True